

PBTA Analysis – Summary Report

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1 Summary

Questions like what cancer group is interesting or what is interesting in the data are challenging questions and could be tackled in few ways. A simple yet powerful approach would be to tackle groups that has strong statistics to work on (number of samples and so on). Picking these groups and conducting an analysis on them would be reproducible among other groups and can reveal novel outcomes.

Never the less, to answer these questions I conducted a systematic "unsupervised" approach where I assumed no prior knowledge, or bias on the data. I wanted to highlight interesting variables (clinical, genomic) that can shed light on cancer groups/sub groups (CG, CSG) in the data. To do so i first investigated the clinical metadata by visualizing and statistically questioning is variable X ("Age", "Sex", "Tf", etc.) differs in cancer groups and is it related to death/event status. I evaluated X combinations and found that: cancer groups yyy, differs in variables „,. I further conducted unsupervised clustering techniques on these variables to explore the data structure and found: Second, I utilized the mRNA expression matrix and tried to identify correlated, mutual exclusive, and co-occurring genes. I further tried to connect this insights into my previous findings on the clinical data and reveal novel sub groups among patients. I found that the genes: X,Y,Z are in.. A further inspection about the role of these genes, their biological pathways is needed.

2 Introduction

3 Methods

3.1 Data processing

I processed the clinical metadata of the PBTA RNA study which includes two tables: patient-level and sample-level tables with 2870 and 4312 samples, respectively. I merged the two tables on the patient id, which resulted in 4074 samples (2701 unique patients) with a known CG. To analyze the data I converted the OS and EFS status values into labels and event flags, merged missing RACE and ETHNICITY values into "Unknown", mapped the cancer PREDISPOSITION into "No predisposition"/"Unknown", and merged blank or "To be classified" molecular subtypes into "Unclassified". For the AGE distribution I detected outliers with the Tukey IQR rule ($AGE > Q3 + 1.5 * IQR = 27.125 \text{ yr}$), which flagged 62 patients.

3.1.1 Statistic analysis

To evaluate the data statistically, I computed the % of missing values in each column to identify noisy variables. To compare CGs, I tested every CG with $n \geq 20$ samples (excluding 33 CGs): SEX was tested against a 50:50 baseline with a binomial test, while PREDISPOSITION and RACE compositions were tested with Chi-squared tests (Fisher exact when the expected

counts were < 5), reporting Cramer's V as the effect size. For the numeric variables AGE, TUMOR_FRACTION (TF), and TUMOR_PLIDY (TP) i used a Kruskal-Wallis test followed by per-group Mann-Whitney tests (each CG vs all the others). i corrected the p-values within each family of tests using the Benjamini-Hochberg FDR method and considered $q < 0.05$ as significant, always reporting the raw p next to the adjusted q. Finaly, i computed spearman (and pearson) correlations between AGE, TF and TP, and ordered the CGs by similar race composition using hierarchical clustering. All the processing and statistics were implemented in python (pandas, scipy, statsmodels) as executed notebooks.

3.1.2 Survival analysis

To test whether the previously analyzed variables affect survival, I used time-to-event analyses on the OS and EFS endpoints. I plotted Kaplan–Meier (KM) curves per CG ($n \geq 20$) and each group's survival was compared against all others combined with log-rank tests (multiple-testing correction as above). Each variable was then tested against survival status (Mann–Whitney U for AGE, TF, and TP; Chi-squared for SEX and predisposition), and numeric variables were dichotomized at their median to test KM separation with log-rank tests. Finally, I fitted a multivariate Cox proportional-hazards model with AGE, SEX, TUMOR_FRACTION, and TUMOR_PLOIDY as covariates, stratified by cancer group, on complete cases, reporting Wald p-values, hazard ratios with 95% confidence intervals, and Schoenfeld residuals to check the proportional-hazards assumption. Analyses were implemented in Python (scipy, lifelines) as executed notebooks.

For individual cancer groups..

4 Results

4.1 Meta data and clinical analysis

To begin with i conducut a simple visualization and summarization of the clinical and demographic data including statistics and histograms. I noticed that event-free survival (EFS) and overall survival (OS) months variables has 28% Null values (Figure 1A). In total 65.3% has a living status via the OS variabel, whereas for the EFS variable, half the patients exhibit no event and the other half splitted between different events (Figure 1B, C). This demonstrate the value of investigating the EFS rather than OS where EFS provides additional information cite. Furthermore, the "AGE" (age of diagnosis) right skewed histogram suggests outliers or misleading data points as this is a kids dataset (Figure 1D). A further inspection revealed that most of this data points (53/62) belong to oligodendrogliomas paitens (Figure 1E). A quick web search suggested that this cancer is an adult cancer, and therefor removing these data points for subsequeant analysis should be consider.

Investigating the samples dataset demonstrated 20-40% Null values in Tumor fraction (TF) and ploidy (TP), RNA selection, and matching id samples (Figure 2A). Among the 55 cancer groups Low-grade glioma (LGG) has most cases (862). 3 groups has 300-500, 7 groups has 70-200 cases, and 45 groups has less then 70 cases (Figure 2B). The molecular subtype information aligns well with the cancer group, despite 13 molecular subtypes belonging to 2 or more disticnt CGs (Figure 2C). This suggests eather wrong labeling of the sub molecular type or some sub type belonging to multiple CGs.

4.2 Cross datasets analysis

Next, to reveal diffrences between cancer groups, I evaluated whether certain variales are enriched among diffnernet cancer groups. To do so, I evaluated the categorical variables "SEX",

"predisposition", and "race" using Binomial, Chi-squared and Fisher exact tests. For the numeric variables "AGE", "TF", and "TP" I used Kruskal-Wallis and per-group Mann-Whitney tests between CGs.

I found that for 6 CGs the male to female distribution is significantly different (Figure 3A) compared to 50:50 baseline. Furthermore, among patients with predispositions (9%), 5 CGs have different predisposition composition (Figure 3B). Lastly, 9 CGs showed difference in race composition compared to the overall race composition in the dataset (Figure 3C). These findings suggest that sex, race and predisposition could co-occur with different CGs. Lastly as demonstrated in (Figure 3C), some CGs show similar predispositions compositions, suggesting a relation between these CGs. A further inspection on these CGs could reveal the connection. Next, I demonstrated that 14, 6, and 4 CGs are significantly different by AGE, TF, and TP variables (Figure 3D, E, F). Investigating their Pearson/Spearman correlations (Figure 3G, H) yielded significant p values (5/6) with no observed trends possibly due to high sample amount.

4.2.1 Survival analysis

To test differences in survival patterns globally and across CGs, I evaluated all the previous analyzed variables and tested their effect on OS and EFS. First I plotted the KM curves for the top CGs and compared each group's survival rates against the rest of the groups (Figure 4A OS, B EFS). I demonstrated that 15 and 16 CGs have statistically different KM plots for OS and EFS, respectively. Next, I evaluated whether each variable differs the survival states using the Mann-Whitney test globally across the data. I found that the variable x,y,z are different in the OS (Figure 4C) where the variables are different in the EFS (Figure 4D). To estimate the global difference for these variables on the KM plot I dichotomized the variable by their median and binary value for numeric and binary variables, respectively. I found that the variables X,y,z differs in OS (Figure 4E) whereas a,b,d differs in EFS (Figure 4F). Lastly, to evaluate the combination of these variables together I applied a Cox-hazard model on the global dataset. For OS, only the AGE variable is significant with 95% CI and HR of (xx - xx, yy) (Figure 4G) meaning higher age decrease the hazard. For EFS both AGE and TP are significant with CIs and HRs of (xx -xx ,yy, zz-zz, kk, respectively) (Figure 4H) meaning higher age decrease the hazard and higher TP increases the hazard. Interestingly when examining the TF effect on OS outcome higher TF values has lower death rate. Contradictly, the higher TF in the KM has lower survival rates. This can be explained by the fact that a lot of samples with high TF values (1) has lower death rate (why?) but at high tf ratios (0.7-0.9) there is high death rate. The TF=1 is might bias the analysis and future analysis should consider excluding TF=1 samples.

Next, I evaluated if these variables differ the OS and EFS outcomes inside each CGs. To do so, for each group, I dichotomized the samples based on the evaluated variables and compared the KM plots. I demonstrated that with OS outcome AGE, TF, TP, PREDISPOSITION, and SEX differs in x, y, z, a, and GC, respectively ???. For EFS outcome AGE, TF, TP, PREDISPOSITION, and SEX differs in x, y, z, a, and GC, respectively ???. To further evaluate the AGE power to differentiate patients inside CG, I divided the AGE variable in 10% increasing ratios and evaluated the differences on the OS and EFS outcomes. I concluded that the differences are more statistically significant (the p-values are much lower) ???. Lastly, I applied the Cox-PH model to every CG and demonstrated that for OS outcome, in GNT tumor higher AGE increases the risk whereas in DMG, HGG, and MB CGs it decreases the risk. Moreover, in both OS and EFS outcomes all examined variables except TP in OS have a significant effect on risk.

5 Conclusions and Future Directions

6 References